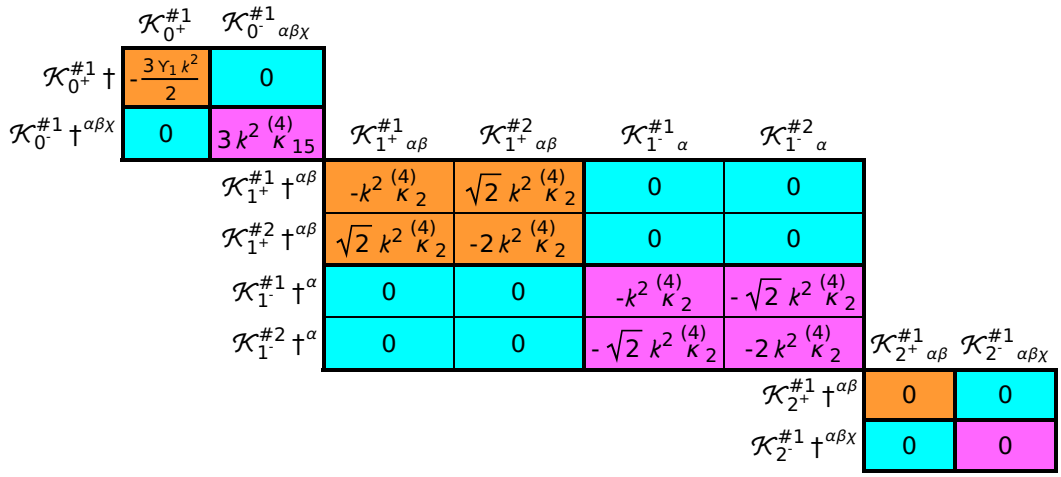

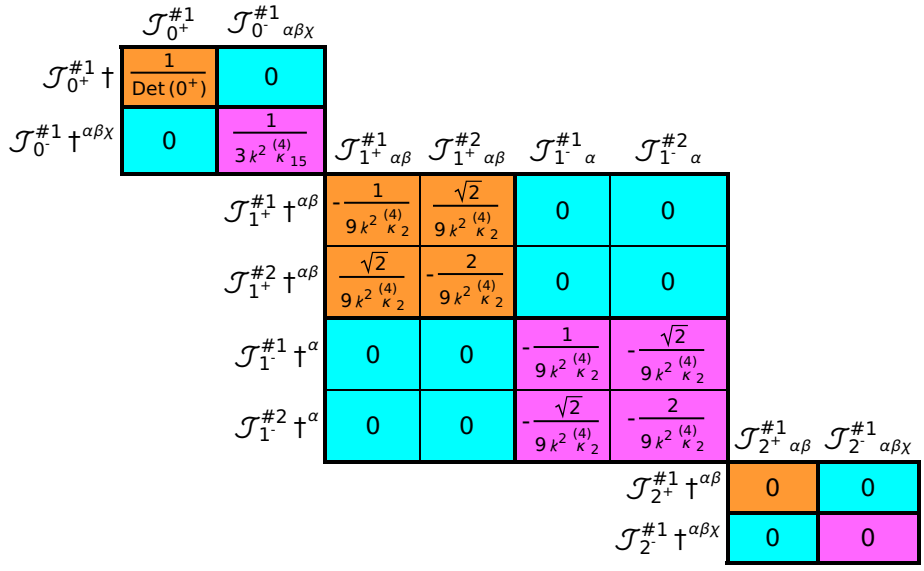

Abbreviations used in matrices

$$Y_1 == {}^{(4)}\kappa_1 + {}^{(4)}\kappa_2 \;\&\&\; \text{Det}(0^+) == -\frac{3}{2}k^2\left({}^{(4)}\kappa_1 + {}^{(4)}\kappa_2\right)$$

Lagrangian

$$\begin{aligned}&{}^{(4)}\kappa_1\,\partial_\beta\mathcal{K}^{\delta}{}_{\chi\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + {}^{(4)}\kappa_2\,\partial_\chi\mathcal{K}^{\delta}{}_{\beta\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + 2\,{}^{(4)}\kappa_{15}\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\alpha\chi}{}^{\delta} + \\ &{}^{(4)}\kappa_2\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\alpha\chi}{}^{\delta} - 4\,{}^{(4)}\kappa_{15}\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta} + 4\,{}^{(4)}\kappa_2\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta} - \\ &2\,{}^{(4)}\kappa_{15}\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta} - {}^{(4)}\kappa_2\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta} - 6\,{}^{(4)}\kappa_2\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\,\partial_\delta\mathcal{K}_{\chi\beta}{}^{\delta} - {}^{(4)}\kappa_{15}\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi} - \\ &2\,{}^{(4)}\kappa_2\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi} + {}^{(4)}\kappa_{15}\,\partial_\delta\mathcal{K}_{\alpha\beta\chi}\,\partial^\delta\mathcal{K}^{\alpha\beta\chi} - 2\,{}^{(4)}\kappa_{15}\,\partial_\delta\mathcal{K}_{\beta\alpha\chi}\,\partial^\delta\mathcal{K}^{\alpha\beta\chi}\end{aligned}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\;\mathcal{J}_{\alpha\beta\chi}$
Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}_{1^-}^{\#1\,\alpha} == \mathcal{J}_{1^-}^{\#2\,\alpha}$	3	$\partial_\chi\partial^\alpha\mathcal{J}^{\beta}{}_{\beta}{}^{\chi} + \partial_\chi\partial^\chi\mathcal{J}^{\beta\alpha}{}_{\beta} == 0$
$2\,\mathcal{J}_{1^+}^{\#1\,\alpha\beta} + \mathcal{J}_{1^+}^{\#2\,\alpha\beta} == 0$	3	$\partial_\chi\mathcal{J}^{\alpha\beta\chi} + \partial_\chi\mathcal{J}^{\chi\alpha\beta} == \partial_\chi\mathcal{J}^{\beta\alpha\chi}$
$\mathcal{J}_{2^-}^{\#1\,\alpha\beta\chi} == 0$	5	$3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta}{}_{\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\alpha\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\beta\epsilon} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\alpha}{}_{\delta} == 3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha}{}_{\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\beta\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\alpha\epsilon} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\beta}{}_{\delta}$
$\mathcal{J}_{2^+}^{\#1\,\alpha\beta} == 0$	5	$3\,\partial_\delta\partial_\chi\partial^\alpha\mathcal{J}^{\chi\beta\delta} + 3\,\partial_\delta\partial_\chi\partial^\beta\mathcal{J}^{\chi\alpha\delta} + 2\,\eta^{\alpha\beta}\,\partial_\epsilon\partial^\epsilon\partial_\delta\mathcal{J}^{\chi}{}_{\chi}{}^{\delta} == 2\,\partial_\delta\partial^\beta\partial^\alpha\mathcal{J}^{\chi}{}_{\chi}{}^{\delta} + 3\,(\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\alpha\beta\chi} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\beta\alpha\chi})$
Total # constraint(s):	16	
Resolved unitarity condition(s):		True